

Research Statement

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Research Agenda

My research examines how information and institutional design shape communication, cooperation, and coordination. I combine formal game-theoretic models with controlled experiments to study when theoretical predictions guide behavior and why they sometimes fail.

A central theme of my work is that institutions and information can affect different participants through distinct strategic, cognitive, and behavioral margins. A punishment that disciplines an informed sender may simultaneously discourage an uninformed receiver from trusting. A failure of collective resistance may arise not from an inability to coordinate, but from uncertainty about whether an observed attack reveals a broader threat. A contract that reduces the theoretical cost of implementing team effort may instead impose greater reasoning demands or generate ex-post inequality. Understanding these specific responses is essential for designing institutions that remain effective beyond their theoretical frameworks.

My job market paper studies the distinct foundations of honesty and trust in repeated communication. Two additional projects examine the informational foundations of collective resistance and the cost and behavioral robustness of teamwork incentives. A complementary project studies how personal and social counterfactuals affect individual choice. Together, these projects form a research agenda on *Communication and Coordination under Incomplete Information: Theory and Experiment*.

Honesty, Trust, and the Shadow of the Future

My job market paper, *Role Uncertainty and Punishment Severity in Repeated Cheap Talk: Theory and Experiment*, coauthored with Wooyoung Lim, asks whether repeated relationships can sustain informative communication when the interests of an informed sender and an uninformed receiver are only partially aligned. The paper has received a Revise and Resubmit decision from *Games and Economic Behavior*.

Motivated by repeated communication in long-term professional relationships, such as those arising in reference-letter systems, investment syndication, and the “revolving door” between the public and private sectors, we study an indefinitely repeated cheap-talk game. In each round, one player privately observes the state and sends a message. The receiver then chooses between two state-contingent actions and an intermediate safe action that yields state-independent payoffs. The sender always prefers one particular action, whereas the receiver prefers to match the action with the state. In the one-shot equilibrium, the receiver ignores the message and chooses the safe action. Repetition can instead support truthful communication through history-dependent strategies.

The model varies two institutional features. The first is *punishment severity*. A harsher punishment reduces the sender’s payoff following a breakdown of communication, but also increases the receiver’s payoff from the safe action. The second is *role uncertainty*. Under fixed roles, participants remain sender and receiver throughout the relationship. Under role uncertainty, they may alternate between the informed and uninformed roles.

Our main theoretical benchmark is a simple truth-telling equilibrium in which the sender reports the state truthfully, the receiver follows the message, and any deviation triggers permanent reversion to the one-shot outcome. Each institutional feature can facilitate this equilibrium when introduced separately. With fixed roles, harsher punishment strengthens the sender’s incentive to be honest. Under mild punishment, role uncertainty allows both participants to share the informational surplus from occupying the receiver role.

Their interaction is more subtle. Under role uncertainty, punishment severity becomes a double-edged instrument: it reduces a player’s punishment payoff when that player is the sender but raises it when the same player is the receiver. With equally likely roles, these effects cancel. With unequal role probabilities, the binding incentive constraint can shift across players as punishment becomes more severe. Consequently, moderate punishment may facilitate truthful communication, whereas excessively harsh punishment may

hinder it.

We test these predictions in six treatments that cross mild or harsh punishment with fixed, unequal-chance random, or equal-chance random roles. We separately measure communication efficiency, sender truthfulness, and receiver trust. This decomposition is important: high efficiency need not imply that information is truthfully transmitted, and greater honesty need not improve communication efficiency if receivers remain unwilling to trust.

The current evidence reveals three main patterns. First, harsh punishment with fixed roles disciplines senders but does not improve efficiency. Senders become more truthful, yet receivers do not respond with greater trust. Instead, they increasingly select the intermediate action, whose payoff is made especially attractive by the same parameter that punishes the sender. We call this the *safety trap*: an instrument designed to solve the sender's incentive problem simultaneously increases the receiver's opportunity cost of trust.

Second, role uncertainty under mild punishment increases trust without producing a comparable increase in honesty. This pattern is consistent with a *Blind Trust Equilibrium*: senders always send the message that induces their preferred action, while receivers follow that message without extracting information from it. Role rotation allows participants to take turns benefiting from this permissive behavior. The resulting outcome can be efficient under the paper's surplus criterion even though the messages themselves are uninformative.

Third, combining harsh punishment with role uncertainty leads to a larger decline in trust despite somewhat greater sender honesty. One interpretation is that the higher safe payoff discourages receivers from acting on messages. Participants may therefore behave honestly when acting as senders but conservatively when acting as receivers, separating their behavior across roles. Within-game evidence reinforces this interpretation: receivers often choose the safe action from the first round or abandon established cooperation without a preceding lie, rather than using it primarily as a conditional punishment.

These findings distinguish honesty, trust, and efficiency. Truthful communication is a relatively high-tension form of cooperation: it requires disciplined senders and receivers who are sufficiently confident to act on their reports. Blind trust is a lower-tension form of cooperation that reduces the need for verification but tolerates dishonesty. The preferred institution therefore depends on whether the objective is to maximize surplus or to elicit genuinely informative reports.

The discrepancies between our theoretical predictions and experimental findings also motivate a broader question: what are the boundaries of the observed *safety trap* and *Blind Trust Equilibrium*? Promising approaches include separating sender sanctions from the receiver's safe payoff and examining whether reducing the sender's payoff from the preferred action when interests conflict can make a truth-telling equilibrium Pareto-superior to a blind-trust equilibrium in the modified game.

Other Current Research

The Informational Foundations of the Failure of Collective Resistance

In *First They Came for the Others: A Theory of Divide-and-Conquer*, coauthored with Yeon-Koo Che and Wooyoung Lim, we ask why potential victims fail to resist an aggressor collectively even when joint resistance would be effective. Standard explanations emphasize conflicting interests, free riding, or mechanical coordination failures. We instead identify an epistemic friction: after observing an attack on an initial target, a bystander must infer whether it reflects a target-specific motive or the beginning of a systemic campaign.

An attacker privately observes the benefits of conquering two potential victims and may attack them sequentially. The first victim decides whether to resist. If she resists, the second victim observes the attack and the resistance decision and chooses whether to join. This timing removes the standard simultaneous-move coordination problem. If the second victim knew that she would also be targeted, she would join the first victim's resistance; anticipating this support, the first victim would fight. Within the model, any

failure of collective resistance therefore stems from uncertainty about the attacker’s target scope.

The model produces two canonical equilibrium outcomes. In a divide-and-conquer equilibrium, the initial attack is sufficiently consistent with a localized motive that the bystander’s posterior probability of being targeted remains below her danger threshold. She stays out; the first victim, anticipating solitary resistance, surrenders; and the attacker can proceed sequentially. In a collective-resistance equilibrium, the same attack is interpreted as evidence of a broader campaign. The bystander joins, the first victim fights, and only attacker types willing to face joint resistance initiate an attack. Thus, the equilibrium outcome turns on a single inferential question: after observing the first attack, how likely does the bystander believe she is to be next?

The first comparative static generated by this posterior logic is counterintuitive. A higher cost of aggression can make divide-and-conquer easier to sustain because, conditional on an attack occurring, the bystander is more likely to rationalize it as arising from an exceptional motive against the first victim. Second, lower correlation between the victims’ values to the attacker weakens the inference that their fates are linked. The greater perceived specialness of the first target makes the attack appear more localized. We further show how wishful and magical thinking, particularizing rhetoric, neutrality and collective-defense treaties, and downstream alliance networks reshape this inference. The central implication is that effective collective defense requires not only the ability to coordinate resistance, but also a common interpretation of what the first attack reveals.

Designing Cost-Effective Incentives for Teamwork

In *Incentivizing Teamwork*, coauthored with Timothy Cason, we study how a principal can induce two workers to exert costly, complementary effort when only aggregate project output is contractible. The principal offers outcome-contingent bonuses and chooses an information structure along two dimensions: whether effort decisions are simultaneous or sequential, and whether compensation is public or privately mediated. These dimensions capture two distinct forms of organizational transparency: the observability of coworkers’ effort and the informativeness of their compensation.

The design compares a symmetric-bonus benchmark with four mechanisms that uniquely implement joint effort through iterated dominance. The benchmark makes joint effort a Nash equilibrium but does not uniquely implement it. Under simultaneous public contracting, asymmetric bonuses initiate the dominance chain. The simultaneous-mediated mechanism instead uses uncertainty about bonus ranks to reduce cost and restore ex-ante pay-symmetry. With sequential public decisions, the later worker’s response creates endogenous peer monitoring. The sequential-mediated mechanism jointly randomizes move order and bonuses, eliminating the late-mover premium while preserving ex-ante symmetry. Across the five schemes, theoretical compensation costs satisfy the strict ranking: simultaneous public, simultaneous mediated, symmetric benchmark, sequential public, and sequential mediated, from highest to lowest.

The planned experiment will test whether these cost savings survive cognitive and distributional frictions. Standard theory predicts joint effort under all four uniquely implementing mechanisms, so lower compensation costs translate into greater cost-effectiveness under this benchmark. In practice, mediation may increase reasoning demands and create ex-post inequality despite ex-ante symmetry, while sequential play may facilitate peer monitoring but also activate reciprocity or spite. Our Fehr–Schmidt analysis identifies the sequential-public mechanism as particularly robust to inequity aversion and shows how the theoretically cheapest sequential-mediated mechanism can lose its cost advantage when realized inequality discourages effort. A simulation-based power analysis informs the pre-registration and planned data collection.

Social Counterfactuals and Correlation Sensitivity

In *Correlation Sensitivity in Social Context: Theory and Experiment*, coauthored with Hendrik Rommeswinkel, we study how people evaluate social allocations using two types of counterfactuals: unrealized personal alternatives and social differences involving others. Our framework distinguishes regret over forgone personal alternatives from social regret over forgone social differences, thereby capturing behavior associated with fear of missing out. We are developing an experimental task that separately

identifies these channels while holding the relevant marginal distributions over allocations constant.

Future Research

My future research will examine how institutional and informational interventions affect strategic behavior: through which channels do they achieve their intended effects, and when are those effects redirected or offset by participants' responses? Building on my current work, I will study how changes in incentives interact with beliefs, strategic reasoning, and social evaluations. My goal is to identify the mechanisms behind these interactions and derive testable predictions about when findings from one strategic environment should extend to another.

A near-term extension of my job market paper will separate sender sanctions from the receiver's safe payoff. This design will test whether stronger discipline produces more informative communication when it no longer makes the receiver's alternative to trusting more attractive. Combining these treatments with belief elicitation and separate measures of honesty and trust will help distinguish changes in sender behavior from changes in receivers' expectations and willingness to act on messages. The broader question is when relaxing one participant's incentive constraint improves cooperation, and when the resulting outcome remains limited by another participant's response.

I will also investigate interventions that change the informational meaning of strategic actions. Our divide-and-conquer experiment, currently being prepared for pre-registration, varies the cost of aggression and the prior probability of a systemic attacker to test the inference channel underlying collective resistance. Additional phases examine propaganda through assurances of safety with different costs of breaking one's word, and pre-game histories through provocation and disclosure of prior trust-game choices. Belief elicitation and conditional decisions will help examine how these features affect perceived threats and expectations of support. This research will clarify when changes in the attacker's incentives and the victims' beliefs reinforce or counteract one another in shaping collective resistance.

A complementary direction will distinguish barriers that call for different institutional remedies. In teamwork, low effort under a theoretically effective contract could reflect difficulty reasoning through others' choices or an unwillingness to exert effort under unequal compensation. I plan to vary reasoning support and payoff inequality separately, preserving the relevant material incentive comparisons. These tests will examine whether simplifying strategic reasoning restores the effectiveness of an incentive scheme, or whether its distributional consequences remain an obstacle. The aim is to explain not only why a mechanism underperforms, but which modifications address the underlying friction.

Taken together, these projects will contribute to a broader understanding of institutional effectiveness: when changes in incentives and information produce their intended outcomes, and when they leave other obstacles intact or create new ones. By linking formal predictions to experimental evidence on participants' responses, I aim to explain how a design's incentive structure interacts with participants' beliefs, reasoning, and responses to payoff inequality. The broader value of these explanations will lie in identifying the conditions under which an intervention success depends. This provides a basis for extending insights beyond a particular game without assuming that the same design will work everywhere. My goal is to contribute to behavioral game theory and institutional design with explanations that guide both the evaluation of interventions and their adaptation to new strategic environments.